

SINGULARITIES ON NORMAL VARIETIES

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ABSTRACT. In this paper we generalize the definitions of singularities of pairs and multiplier ideal sheaves to pairs consisting of formal linear combinations of proper closed subschemes of an arbitrary normal variety X . No assumption of X being $\mathbb{Q}$ -Gorenstein or the pair being log $\mathbb{Q}$ -Gorenstein is required. Among other things, we show that the following fundamental and well known results extend to this more general setting: Nadel's vanishing theorem, Shokurov–Kollár's connectedness principle, Elkik's theorem on rational singularities, Kawamata's subadjunction theorem, Siu's invariance of plurigenera for varieties of general type, Nakayama's invariance of numerical Kodaira dimension, and Kawamata's invariance of canonical singularities.

INTRODUCTION

The theory of singularities of pairs and multiplier ideals has become a core part of the study of higher dimensional algebraic varieties, with virtually no limit to applications (e.g., see [Kol2, KM, Laz, EM2] for an overview of the theory and various applications).

The main purpose of this paper is to investigate possible extensions of the theory to settings which are more general than the ones in which it has been defined and studied. Our priority, naturally, is to perform such generalization in such a way that most of the essential features are preserved.

Framework. Even if one is only interested in the actual singularities of a given variety X , it is often convenient (and sometimes necessary) to work in a more general framework, by considering *pairs* (X, Z) , where Z is a formal linear combinations of proper subschemes of X . Pairs naturally arise in a geometric meaningful way in a variety of instances: as boundaries of open varieties, markings on varieties in moduli problems, discriminants and orbifold structures of morphisms, base schemes of rational maps, and inductive tools in higher dimensional geometry.

When the canonical class K_X is $\mathbb{Q}$ -Cartier, several invariants have been defined via resolution of singularities. A key ingredient in their definitions is the relative canonical divisor of a resolution $f: Y \rightarrow X$, that is, the unique exceptional divisor supported on the exceptional locus of f and numerically equivalent to the difference $K_Y - f^*K_X$. Its role corresponds that of the Jacobian in the change-of-variable formula for integrals. The difficulty in extending the definitions of such invariants to an arbitrary (normal) variety X arises as soon as K_X is not $\mathbb{Q}$ -Cartier, as it is unclear in this case what should be its pullback. One way around the problem is to perturb K_X by adding a *boundary* divisor Δ so that $K_X + \Delta$ is $\mathbb{Q}$ -Cartier. This also gives rise to a pair (X, Δ) , but the boundary itself

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may not have any particular geometric meaning (in fact the choice of Δ is in general not canonical, and different choices may give rise to different classes of singularities).

Main results. Our approach to the problem is different and more direct. We develop a notion of divisorial valuation and pullback of (Weil) $\mathbb{Q}$ -divisors which agrees with the usual one for $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisors. In this way we are able to define the *relative canonical divisor* $K_{Y/X}$ for any proper birational morphism $f: Y \rightarrow X$ of normal varieties, as the unique exceptional $\mathbb{R}$ -divisor numerically equivalent to $K_Y + f^*(-K_X)$. We also define a suitable ‘approximation’ of this $\mathbb{R}$ -divisor via $\mathbb{Q}$ -divisors $K_{m,Y/X}$ (for $m \geq 1$), that we call *limiting relative canonical divisors*. Quite interestingly, it turns out that there is a potential dichotomy in the theory of singularities (which disappears in the $\mathbb{Q}$ -Gorenstein case), corresponding to the expected phenomenon for which such limiting divisors $K_{m,Y/X}$ may fail to converge to $K_{Y/X}$ (as $m = k!$ for integers $k \rightarrow \infty$) if K_X is not $\mathbb{Q}$ -Cartier.

Using these notions of ‘relative canonical divisors’, we generalize the definitions of multiplier ideals and singularities of pairs to pairs of the form (X, Z) , where X is an arbitrary normal variety and $Z = \sum b_k \cdot Z_k$ is a formal linear combination of proper closed subschemes of X . All the main features are preserved in this generalization.

Our definition of *canonical* and *terminal singularities* uses log discrepancy conditions with respect to the relative canonical divisor $K_{Y/X}$. The main features of canonical singularities, such as the deformation invariance properties of plurigena (for singular varieties of general type), of numerical Kodaira dimension and of canonical singularities established in works of Siu, Nakayama and Kawamata, also hold in our more general setting (see Theorems 9.4 and 9.5).

By contrast, in order to generalize the notions of *log terminal* and *log canonical singularities*, we impose log discrepancy conditions with respect to the limiting relative canonical divisors $K_{m,Y/X}$. This allows us to preserve the main features of this side of the theory. For instance, we obtain that normal varieties with log terminal singularities (as defined in this generality) have rational singularities (see Theorem 8.5), generalizing to our setting the well-known result of Elkik. We also show that the minimal log canonical centers studied by Kawamata (which in general are not known to be $\mathbb{Q}$ -Gorenstein) are log terminal (see Theorem 8.7); this provides in particular a natural setting for the theory developed in this paper. In fact, the subadjunction theorem itself is generalized to our context (see Theorem 8.10). Finally, we check that in dimension two our notions of log terminal and log canonical singularities agrees with the one of numerically log terminal and numerically log canonical singularities (see Corollary 8.12 and Proposition 8.13), which in particular implies that they are always $\mathbb{Q}$ -Gorenstein.

We expect that the larger freedom in defining these notions of singularities should have interesting applications. In the (log-) $\mathbb{Q}$ -Gorenstein setting many applications rely on multiplier ideals and their vanishing theorems, and it is encouraging that these powerful methods extend to our setting. The *multiplier ideal sheaf* of a pair (X, cZ) is defined, in our generality, as the unique maximal element in the collection of ideal sheaves

$$\{(f_m)_* \mathcal{O}_{Y_m}(\lceil K_{m,Y_m/X} - c \cdot f_m^{-1}(Z) \rceil)\}_{m \geq 1},$$

where for every m the morphism $f_m: Y_m \rightarrow X$ is a suitably ‘high enough’ log resolution of (X, Z) depending on m . An important result we can prove is that these multiplier ideal sheaves still satisfy the expected vanishing properties (see Theorem 5.1). As an application, we deduce a generalization of the connectedness principle of Shokurov and

Kollár (see Corollary 5.5). We also generalize the definition of adjoint ideal sheaves, and obtain a basic inversion-of-adjunction statement (see Corollary 6.8).

Throughout the paper, we compare our definitions with the corresponding ones given in the more familiar case in which a boundary Δ such that $K_X + \Delta$ is $\mathbb{Q}$ -Cartier is fixed. In particular, we show that in general multiplier ideal sheaves can be described as a certain adjoint ideal sheaves of the pair resulting by adding a suitable boundary (see Proposition 6.11).

Motivations. The original motivation of this research comes from a question posed by Valery Alexeev during the AIM Workshop [AIM06], which asks whether it is possible to generalize the definitions of singularities of pairs in a wider context than the usual one. The question itself was motivated by an example, due to Paul Hacking, of a flat family of pairs (S_t, D_t) , where S_t is a smooth surface and D_t is an effective divisor, that specializes to a pair (S_0, D_0) , where S_0 is a singular surface and the ideal sheaf of D_0 acquires an embedded prime at the singularity of S_0 .

The example brings to light an important issue: namely, that often, in the literature, pairs (X, Z) have been intended in a combined way, both geometrically (as in one of the situations previously described) and as a correction to the possible failure of K_X being $\mathbb{Q}$ -Cartier—by incorporating a boundary Δ into Z , so to speak. We insist in this paper to keep the two things separated. To this end, we introduce a new terminology, calling *log variety* any normal variety X equipped with a boundary Δ for which $K_X + \Delta$ is $\mathbb{Q}$ -Cartier. Such an object will be denoted by X^Δ ; the ‘pair’ given by a log variety X^Δ together with a formal linear combination $Z = \sum b_k \cdot Z_k$ will be denoted by (X^Δ, Z) , and called *log pair*.

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1. NOTATION AND CONVENTIONS

We work in the category of algebraic varieties; in particular, all morphisms are implicitly intended to be algebraic morphisms of algebraic varieties. Starting from Section 4, we will restrict the setting to varieties defined over an algebraically closed field of characteristic zero.

Let X be a normal algebraic variety. Throughout the paper, a *divisor* on X will be understood to be a Weil divisor, unless otherwise specified. The group of divisors is denoted by $\text{Div}(X)$. If ϕ is a non-trivial rational function on X , then we denote by $\text{div}(\phi)$ the principal divisor associated to it. For $L \in \{\mathbb{Q}, \mathbb{R}\}$, an element of $\text{Div}(X) \otimes L$ is called a *L -divisor*. $\mathbb{Q}$ -linear equivalence between $\mathbb{Q}$ -divisors is denoted by $\sim_{\mathbb{Q}}$. For two L -divisors A and B , we write $A \geq B$ if $A - B$ is effective, and denote by $\lceil A \rceil$ and $\lfloor A \rfloor$ the componentwise round up and round down of A . A $\mathbb{Q}$ -divisor D on X is said to be *$\mathbb{Q}$ -Cartier* if there exists a positive integer m such that mD is a Cartier divisor. Recall that X is said to be *$\mathbb{Q}$ -factorial* if every divisor is $\mathbb{Q}$ -Cartier.

For a closed subscheme $Z \subseteq X$, we denote by $\mathcal{I}_Z \subseteq \mathcal{O}_X$ its ideal sheaf. Let $K = K(X)$ be the function field of X , and let $\mathcal{K} = \mathcal{K}_X$ be constant sheaf of rational functions on X . We call (*coherent*) *fractional ideal sheaf* any subsheaf $\mathcal{I} \subset \mathcal{K}$ that is (locally) finitely generated as a sub- $\mathcal{O}_X$ -module of $\mathcal{K}$. For instance, every coherent sheaf of ideals is a fractional ideal sheaf via the inclusion $\mathcal{O}_X \subset \mathcal{K}$, and so is the reflexive sheaf $\mathcal{O}_X(-D) \subset \mathcal{K}$

¹Other people to thank?

associated to any divisor D on X . Note that a fractional ideal sheaf $\mathcal{I}$ is locally generated (as a sub- $\mathcal{O}_X$ -module) by a single element if and only if it is an invertible sheaf, i.e., $\mathcal{I} = \mathcal{O}_X(-D)$ for some Cartier divisor D on X . When X is quasi-projective, a subsheaf $\mathcal{F} \subset \mathcal{K}$ is a fractional ideal if and only if there exists a Cartier divisor C on X such that $\mathcal{O}_X(C) \cdot \mathcal{F} \subseteq \mathcal{O}_X$ is a coherent ideal sheaf.

2. VALUATIONS OF $\mathbb{Q}$ -DIVISORS

Let X be a normal variety. Let $K = K(X)$ be the function field of X , and let $\mathcal{K} = \mathcal{K}_X$ be constant sheaf of rational functions on X .

Definition 2.1. A *divisorial valuation* v on X is a discrete valuation of the form

$$v = q \operatorname{val}_F: K^* \rightarrow \mathbb{Z},$$

where $q \in \mathbb{Z}_+$ and F is a prime divisor on a normal variety X' with a given birational morphism $\mu: X' \rightarrow X$. The valuation is defined by $v(\phi) := q \operatorname{ord}_F(\phi \circ \mu)$ for every $\phi \in K^*$, where ord_F denotes the order at the generic point of F . The image $\mu(F)$ of F in X is called the *center* of v in X , and denoted by $c_X(v)$ (or $c_X(F)$).

Throughout this section, we fix a divisorial valuation v of X . If D is a Cartier divisor on X , then the valuation $v(D)$ of D is given by q times the coefficient of F in the divisor μ^*D . Of course this is compatible with the previous definition: if $\psi \in K^*$ is a rational function defining D in an open set $U \subseteq X$ intersecting the center $c_X(v)$, then $v(D) = v(\psi)$. The valuation $v(Z)$ of a proper closed subscheme $Z \subset X$ is given by

$$v(Z) = v(\mathcal{I}_Z) := \min\{v(\phi) \mid \phi \in \mathcal{I}_Z(U), U \cap c_X(v) \neq \emptyset\}.$$

This definition extends to formal linear combinations $\sum a_k \cdot Z_k$ of proper closed subschemes $Z_k \subset X$ by setting $v(\sum a_k \cdot Z_k) := \sum a_k \cdot v(Z_k)$.

One can generalize the above definitions to arbitrary non-zero fractional ideal sheaves in the following natural way.

Definition 2.2. The *valuation* $v(\mathcal{I})$ of a non-zero fractional ideal sheaf $\mathcal{I} \subset \mathcal{K}$ along v is given by

$$v(\mathcal{I}) := \min\{v(\phi) \mid \phi \in \mathcal{I}(U), U \cap c_X(v) \neq \emptyset\}.$$

More generally, the *valuation* $v(I)$ of a formal linear combination $I = \sum a_k \cdot \mathcal{I}_k$ of fractional ideal sheaves $\mathcal{I}_k \subset \mathcal{K}$ along v is defined by $v(I) := \sum a_k \cdot v(\mathcal{I}_k)$.

Remark 2.3. If $\mathcal{I}$ and $\mathcal{I}'$ are fractional ideal sheaves on X , with $\mathcal{I} \subseteq \mathcal{I}'$, then $v(\mathcal{I}) \geq v(\mathcal{I}')$.

In the case of ideal sheaves, this definition coincides with the one previously given, and if D is a Cartier divisor, then $v(D) = v(\mathcal{O}_X(-D))$. Note that for every fractional ideal sheaf $\mathcal{I} \subset \mathcal{K}$ and every Cartier divisor D on X , we have $v(\mathcal{O}_X(-D) \cdot \mathcal{I}) = v(D) + v(\mathcal{I})$. In particular one can always reduce to compute the valuation of any fractional ideal sheaf as the sum of valuations of a Cartier divisor and of an ideal sheaf.

Consider now an arbitrary divisor D on X .

Definition 2.4. The $\natural$ -*valuation* (or *natural valuation*) along v of a divisor D on X is

$$v^\natural(D) := v(\mathcal{O}_X(-D)).$$

Clearly, we have $v^\natural(D) = v(D)$ for any Cartier divisor D . In general the $\natural$ -valuation of divisors is not linear with respect to the group structure of $\operatorname{Div}(X)$, as the next example shows.

Example 2.5. Let $X = \{xy = z^2\} \subset \mathbb{C}^3$, and let $v = \text{val}_E$, where E is the exceptional divisor of the blow up of X at the origin. Then, for any two lines $L, M \subset X$ (passing through the origin), we have $v^\natural(L) = v^\natural(M) = v^\natural(L + M) = 1$, and thus $v^\natural(L + M) \neq v^\natural(L) + v^\natural(M)$. In particular, $v^\natural(2L) = v^\natural(L)$. Note also that $v^\natural(-L) = 0$.

The lack of linearity is the price one has to pay to get a notion of valuation for divisors which extends the usual notion for Cartier divisors.

Lemma 2.6. *Let C be a Cartier divisor on X . Then $v^\natural(C + D) = v(C) + v^\natural(D)$ for every divisor D on X .*

Proof. Since $\mathcal{O}_X(-C)$ is locally generated by one rational function, one can check that $\mathcal{O}_X(-C - D) = \mathcal{O}_X(-C) \cdot \mathcal{O}_X(-D)$, hence the assertion follows by Lemma ??.

Definition 2.7. To any non-trivial fractional ideal sheaf $\mathcal{J}$ on X , we associate the divisor

$$\text{div}(\mathcal{J}) := \sum_{E \subset X} \text{val}_E(\mathcal{J}) \cdot E,$$

where the sum is taken over all prime divisors E on X . Equivalently, $\text{div}(\mathcal{J})$ is the divisor on X for which $\mathcal{O}_X(-\text{div}(\mathcal{J})) = \mathcal{J}^{**}$. In particular, $\text{div}(\mathcal{O}_X(-D)) = D$ for any divisor D . We call $\text{div}(\mathcal{J})$ the *divisorial part* of $\mathcal{J}$.

Consider now a birational morphism $f: Y \rightarrow X$ from a normal variety Y .

Definition 2.8. For any divisor D on X , the $\natural$ -pullback (or *natural pullback*) of D on Y is given by

$$f^\natural D := \text{div}(\mathcal{O}_X(-D) \cdot \mathcal{O}_Y).$$

In other words, $f^\natural D = \sum \text{val}_E^\natural(D) \cdot E$, where the sum is taken over all prime divisors E on Y .

Lemma 2.6 immediately implies the following additivity property for $\natural$ -pullbacks.

Lemma 2.9. *Let C be a Cartier divisor on X . Then $f^\natural(C + D) = f^*C + f^\natural D$ for every divisor D on X . In particular, $f^\natural C = f^*C$.*

The $\natural$ -pullback behaves reasonably well for composition of morphisms.

Lemma 2.10. *Let $f: Y \rightarrow X$ and $g: V \rightarrow Y$ be two birational morphisms of normal varieties. Then, for every divisor D on X , the divisor $(fg)^\natural D - g^\natural(f^\natural D)$ is effective and g -exceptional. Moreover, if $\mathcal{O}_X(-D) \cdot \mathcal{O}_Y$ is an invertible sheaf, then $(fg)^\natural D = g^\natural(f^\natural D)$.*

Proof. By Lemma 2.9, we have

$$(fg)^\natural(C + D) - g^\natural(f^\natural(C + D)) = (fg)^\natural D - g^\natural(f^\natural D)$$

for every Cartier divisor C . Therefore, after restricting to a quasi-projective cover and replacing D with $C + D$ for some Cartier divisor $C \geq D$, we may assume without loss of generality that D is effective. Then it suffices to observe that $\mathcal{O}_X(-D) \cdot \mathcal{O}_Y \subseteq \mathcal{O}_Y(-f^\natural D)$, with equality holding at the generic point of every codimension one subvariety of Y . For the last assertion, we first remark that the condition that $\mathcal{O}_X(-D) \cdot \mathcal{O}_Y$ is an invertible sheaf remains unchanged if we multiply $\mathcal{O}_X(-D)$ by an invertible sheaf $\mathcal{O}_X(-C)$, and that $\mathcal{O}_X(-D) \cdot \mathcal{O}_Y = \mathcal{O}_Y(-f^\natural D)$ if $\mathcal{O}_X(-D) \cdot \mathcal{O}_Y$ is locally principal.

A basic property of valuations of Cartier divisors is that $v(mD) = m \cdot v(D)$ for every Cartier divisor D on X and every $m \in \mathbb{Z}$. Similarly, if $f: Y \rightarrow X$ is a birational morphism from a normal variety Y , then $f^*(mD) = m \cdot f^*D$ for any such D and m . Thanks to these simple properties, it is possible (and very natural) to extend the definitions of valuation and pullback by setting

$$(1) \quad v(D) := \frac{v(mD)}{m} \quad \text{and} \quad f^*D := \frac{f^*(mD)}{m}$$

for any $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor D on X , where m is any non-zero integer such that mD is a Cartier divisor.

In general, however, $\mathfrak{k}$ -valuations and $\mathfrak{k}$ -pullbacks of arbitrary divisors do not enjoy a similar homogeneity property, as pointed out in Example 2.5. In fact, the case of a line on a quadric cone is an example of a $\mathbb{Q}$ -Cartier divisor D for which $v^{\mathfrak{h}}(D) \neq v(D)$, where the valuation on the right side is intended as defined above. We solve this problem by giving relevance to the asymptotic nature of the definitions given in (1) for $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisors.

Lemma 2.11. *For every divisor D on X and every $m \in \mathbb{Z}_+$, we have $m \cdot v^{\mathfrak{h}}(D) \geq v^{\mathfrak{h}}(mD)$, and*

$$\inf_{k \geq 1} \frac{v^{\mathfrak{h}}(kD)}{k} = \liminf_{k \rightarrow \infty} \frac{v^{\mathfrak{h}}(kD)}{k} = \lim_{k \rightarrow \infty} \frac{v^{\mathfrak{h}}(k!D)}{k!} \in \mathbb{R}.$$

Proof. If $\phi_1, \dots, \phi_m \in \mathcal{O}_X(-D)(U)$ for some open set $U \subseteq X$, then $\text{div}(\phi_i) \geq D$ on U for each i , and therefore $\text{div}(\prod \phi_i) \geq mD$ on U , which means that $\prod \phi_i \in \mathcal{O}_X(-mD)(U)$. This implies that $\mathcal{O}_X(-D)^m \subseteq \mathcal{O}_X(-mD)$, and thus that $m \cdot v(\mathcal{O}_X(-D)) \geq v(\mathcal{O}_X(-mD))$, which proves the first assertion of the lemma. Both equalities in the display of the lemma follow from this inequality, and the fact that the limit is a real number can be seen by reducing to the case of an effective divisor D . $\square$

Definition 2.12. Let D be a $\mathbb{Q}$ -divisor on X . The *valuation* along v of D is

$$v(D) := \lim_{k \rightarrow \infty} \frac{v^{\mathfrak{h}}(k!D)}{k!} \in \mathbb{R}.$$

If $f: Y \rightarrow X$ is a birational morphism from a normal variety Y , then the *pullback* of D on Y is given by

$$f^*D := \sum \text{val}_E(D) \cdot E,$$

where the sum is taken over all prime divisors E on Y .

Within the notation introduced in the definition, note that $k!D$ is a divisor for all $k \gg 0$, so the argument of the limit on the right hand side is well-defined. By Lemma 2.11, the limit itself is well-defined and finite.

Proposition 2.13. *Let D be a $\mathbb{Q}$ -divisor on X , let v be a divisorial valuation on X , and let $f: Y \rightarrow X$ be a birational morphism from a normal variety Y . If D is $\mathbb{Q}$ -Cartier, then the definition of valuation $v(D)$ and of pullback f^*D given in Definition 2.12 agrees with the one given in (1). In general, if D is an arbitrary $\mathbb{Q}$ -divisor, then*

$$v(C + D) = v(C) + v(D) \quad \text{and} \quad f^*(C + D) = f^*C + f^*D$$

for any $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor C on X .

Proof. For clarity, we momentarily denote by $v'(D)$ the valuation of a $\mathbb{Q}$ -divisor D as defined in Definition 2.12. Let C and D be $\mathbb{Q}$ -divisors, with C $\mathbb{Q}$ -Cartier. We have $v^{\natural}(mC) = v(mC)$ for every $m \in \mathbb{Z}$ such that mC is a Cartier divisor, and therefore, observing that $k!C$ is a Cartier divisor for every $k \gg 0$, we get

$$v'(C + D) = \lim_{k \rightarrow \infty} \frac{v^{\natural}(k!C + k!D)}{k!} = \lim_{k \rightarrow \infty} \frac{v(k!C) + v^{\natural}(k!D)}{k!} = v(C) + v'(D)$$

by Proposition 2.6. In particular, if D is $\mathbb{Q}$ -Cartier, then we have $v'(D) = v(D)$. The analogous statements on pullbacks follows from these. $\square$

Remark 2.14. By Lemma 2.11, we have $v(D) \leq v^{\natural}(D)$ and $f^*D \leq f^{\natural}D$ for every divisor D on X .

Proposition 2.15. *Let $f: Y \rightarrow X$ and $g: V \rightarrow Y$ be two birational morphisms of normal varieties. Then, for every $\mathbb{Q}$ -divisor D on X , the divisor $(fg)^*D - g^*(f^*D)$ is effective and g -exceptional.*

Proof. By definition, we have

$$(fg)^*D - g^*(f^*D) = \lim_{k \rightarrow \infty} \frac{(fg)^{\natural}(k!D) - g^{\natural}(f^{\natural}(k!D))}{k!},$$

where the limit is taken componentwise, on the coefficients of the (fg) -exceptional divisors. Then the assertion follows from Lemma 2.10. $\square$

3. RELATIVE CANONICAL DIVISORS

We recall that a *canonical divisor* K_X on a normal variety X is, by definition, the (componentwise) closure of any canonical divisor $K_{X_{\text{reg}}}$ of the regular locus X_{reg} of X . Equivalently, if $h: V \rightarrow X$ is a proper birational morphism from a smooth variety V , then every canonical divisor K_X on X is obtained as the push-forward $K_X = h_*K_V$ of a canonical divisor K_V of V . We also recall that X is said to be $\mathbb{Q}$ -Gorenstein if some (equivalently, every) canonical divisor K_X is $\mathbb{Q}$ -Cartier.

Throughout this section, we fix a proper birational morphism

$$f: Y \rightarrow X$$

of normal varieties.

Lemma 3.1. *The push-forward $f_*: \text{Div}(Y) \rightarrow \text{Div}(X)$ gives a bijection between the canonical divisors of Y and those of X . Moreover, if K_Y and K'_Y are two canonical divisors on Y , the $f_*K_Y - f_*K'_Y$ is a principal divisor and $K_Y - K'_Y = f^*(f_*K_Y - f_*K'_Y)$.*

Proof. The first part follows from the definition of canonical divisor. For the second part, we observe that $K_Y - K'_Y = \text{div}(\psi)$ for some $\psi \in K(Y)^*$, since $K_Y - K'_Y$ is linearly trivial on Y_{reg} and $Y \setminus Y_{\text{reg}}$ has codimension ≥ 2 in Y . Then, observing that f is an isomorphism over an open set $U \subseteq X$ whose complement has also codimension ≥ 2 , we conclude that $f_*K_Y - f_*K'_Y = \text{div}(\psi \circ f^{-1})$, and hence $K_Y - K'_Y = f^*(f_*K_Y - f_*K'_Y)$. $\square$

For the remainder of the section, we fix a canonical divisor K_Y on Y , and let $K_X = f_*K_Y$. The standard notion of relative canonical $\mathbb{Q}$ -divisor (given in the case when K_X is $\mathbb{Q}$ -Cartier) admits two generalizations to non $\mathbb{Q}$ -Gorenstein varieties, corresponding to whether one pulls back K_X or $-K_X$. As we keep into consideration what the main features of the theory of singularities of pairs are, and wish to preserve these in our generalization,

it turns out that there are two different sides of the theory, each of which requires a different approach. One can interpret this by saying that there is a potential dichotomy in the theory of singularities of pairs that only manifests itself in the non $\mathbb{Q}$ -Gorenstein case (cf. Remarks 3.5 and 9.6).

When dealing with the generalization of multiplier and adjoint ideal sheaves, as well as of log canonical and log terminal singularities, we will rely on the following notion.

Definition 3.2. For every $m \geq 1$, the m -th limiting relative canonical $\mathbb{Q}$ -divisor $K_{m,Y/X}$ of Y over X is

$$K_{m,Y/X} := K_Y - \frac{1}{m} \cdot f^{\natural}(mK_X).$$

On the contrary, in order to extend the definitions of canonical and terminal singularities, we consider the following definition.

Definition 3.3. The relative canonical $\mathbb{R}$ -divisor $K_{Y/X}$ of Y over X is

$$K_{Y/X} := K_Y + f^*(-K_X).$$

Note that the definitions of $K_{m,Y/X}$ and $K_{Y/X}$ do not depend on the choice of K_Y . Indeed, if K'_Y is another canonical divisor on Y and $K'_X = f_*K'_Y$, then

$$K_Y + f^*(-K_X) = K_Y + f^*(-K_X + K'_X - K'_X) = K'_Y + f^*(-K'_X)$$

by Proposition 2.13 and Lemma 3.1, and a similar argument applies to $K_{m,Y/X}$. The following property is immediate from the definitions.

Proposition 3.4. If X is $\mathbb{Q}$ -Gorenstein, then $K_{Y/X}$ is the usual relative canonical $\mathbb{Q}$ -divisor, and it is equal to $K_{m,Y/X}$ for every $m \geq 1$ such that mK_X is Cartier.

Remark 3.5. It follows by Lemma 2.11 that, taking the limsup of the coefficients of the components of the $\mathbb{Q}$ -divisors $K_{m,Y/X}$, one obtains the $\mathbb{R}$ -divisor $K_{Y/X}^- := K_Y - f^*K_X$. It is easy to see that $K_{Y/X}^- \leq K_{Y/X}$, and of course the two divisors coincide if X is $\mathbb{Q}$ -Gorenstein. However, we expect that in general they may be different.

Lemma 3.6. Let m be a positive integer, and let $f: Y \rightarrow X$ be a proper birational morphism from a normal variety Y such that mK_Y is Cartier and $\mathcal{O}_X(-mK_X) \cdot \mathcal{O}_Y$ is invertible. Then for every proper birational morphism $g: V \rightarrow Y$ from a normal variety V we have

$$K_{m,V/X} = K_{m,V/Y} + g^*K_{m,Y/X},$$

Proof. Note that $f^{\natural}(mK_X)$ is a Cartier divisor on Y , and thus

$$K_{m,V/X} = K_V - \frac{1}{m} \cdot (gf)^{\natural}(mK_X) = K_V - \frac{1}{m} \cdot g^{\natural}(f^{\natural}(mK_X)) = K_V - \frac{1}{m} \cdot g^*(f^{\natural}(mK_X))$$

by Lemmas 2.10 and 2.9. Since mK_Y is Cartier, we also have

$$K_{m,V/Y} = K_V - \frac{1}{m} \cdot g^{\natural}(mK_Y) = K_V - \frac{1}{m} \cdot g^*(mK_Y) = K_V - g^*K_Y,$$

and moreover

$$g^*K_{m,Y/X} = g^*(K_Y - \frac{1}{m} \cdot f^{\natural}(mK_X)) = g^*K_Y - \frac{1}{m} \cdot g^*(f^{\natural}(mK_X)).$$

The lemma follows. $\square$

Similarly, Proposition 2.15 implies the following.

Lemma 3.7. *For any two proper birational morphisms of normal varieties $f: Y \rightarrow X$ and $g: V \rightarrow Y$, we have*

$$K_{V/X} \geq K_{V/Y} + g^* K_{Y/X},$$

and the difference is g -exceptional.

A different approach to deal with varieties that are not $\mathbb{Q}$ -Gorenstein, largely followed in the last decades, is to introduce a ‘boundary’. The trick is to ‘perturb’ a canonical divisor K_X of X to make it $\mathbb{Q}$ -Cartier.

Definition 3.8. A $\mathbb{Q}$ -divisor Δ is a *boundary* on X if $K_X + \Delta$ is a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor for some canonical divisor K_X of X . If Δ is a boundary, then we denote

$$X^\Delta := (X, \Delta),$$

and refer to it as a *log variety* (or *variety with boundary*). If X is $\mathbb{Q}$ -Gorenstein and $\Delta = 0$, then we identify $X = X^0$; in particular, every normal $\mathbb{Q}$ -Gorenstein variety can be viewed as a log variety with empty boundary.

Again by Lemma 3.1, one sees that if $K_X + \Delta$ is a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor (resp., Cartier divisor) for some canonical divisor K_X , then it is so for every canonical divisor K_X .

Remark 3.9. On every quasi-projective normal variety X , one can always find a divisor Δ giving a boundary on X , and in fact one can always choose such boundary to be effective or anti-effective. Indeed, let K_X be a canonical divisor on X , and write $K_X = P - N$, where P and N are effective divisors. Let $\mathcal{H}$ be an invertible sheaf such that $\mathcal{H} \otimes \mathcal{O}_X(-P)$ is globally generated, and let $H \in |\mathcal{H} \otimes \mathcal{O}_X(-P)|$ be a general element. Then the divisor $\Delta := (H - P) + N$ gives an effective boundary on X . The construction of a boundary given by an anti-effective divisor is analogous.

Definition 3.10. Let Δ be a boundary on X , and let Δ_Y be the proper transform of Δ on Y . The *log relative canonical $\mathbb{Q}$ -divisor* of Y^{Δ_Y} over X^Δ is given by

$$K_{Y/X}^\Delta := K_Y + \Delta_Y - f^*(K_X + \Delta) = K_Y + \Delta_Y + f^*(-K_X - \Delta).$$

Note that $K_X + \Delta = f_*(K_Y + \Delta_Y)$. Clearly, if X is $\mathbb{Q}$ -Gorenstein and $\Delta = 0$, then $K_{Y/X}^0 = K_{Y/X} = K_{m,Y/X}$ for all sufficiently divisible $m \geq 1$. Again, it follows by Lemma 3.1 that $K_{Y/X}^\Delta$ depends on Δ but not on the choice of K_X .

Remark 3.11. There is a small difference between the singularity theory resulting from our interpretation of log variety (see later in the paper for the definitions of log pairs and their multiplier ideals and singularities) and similar treatments in the literature (e.g., in [KM]). The fact is that, in the definition of the relative canonical $\mathbb{Q}$ -divisor $K_{Y/X}^\Delta$, of the boundary Δ we only take into account its contribution along the exceptional divisors, disregarding altogether its proper transform (cf. Remark 7.2).

Lemma 3.12. *With the above notation, suppose that $K_Y + \Delta_Y$ is $\mathbb{Q}$ -Cartier, and let $g: Y' \rightarrow Y$ be a proper birational morphism from a normal variety Y' . Then*

$$K_{Y'/X}^\Delta = K_{Y'/Y}^{\Delta_Y} + g^* K_{Y/X}^\Delta.$$

Proof. By the functoriality of pullbacks of $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisors. □

The following proposition gives the comparison between this ‘log approach’ and the notions of relative canonical divisors introduced in this section.

Proposition 3.13. *For every boundary Δ on X , and every $m \geq 1$ such that $m(K_X + \Delta)$ is Cartier, we have*

$$K_{m,Y/X} = K_{Y/X}^\Delta - \frac{1}{m} \cdot f^\natural(-m\Delta) - \Delta_Y \quad \text{and} \quad K_{Y/X} = K_{Y/X}^\Delta + f^*\Delta - \Delta_Y.$$

Proof. We have $f^*(-K_X) = f^*(-K_X - \Delta + \Delta) = -f^*(K_X + \Delta) + f^*\Delta$ by Proposition 2.13, and similarly $f^\natural(mK_X) = f^\natural(m(K_X + \Delta) - m\Delta) = m \cdot f^*(K_X + \Delta) + f^\natural(-m\Delta)$ by Lemma 2.9. $\square$

Proposition 3.14. *Let Δ and Γ be two boundaries on X with $\Delta \geq 0$ and $\Gamma \leq 0$. Let m be any positive integer such that $m(K_X + \Delta)$ is a Cartier divisor, and let q be any positive integer. Then*

$$K_{Y/X}^\Delta \leq K_{m,Y/X} \leq K_{mq,Y/X} \leq K_{Y/X} \leq K_{Y/X}^\Gamma.$$

Proof. The inequalities $K_{m,Y/X} \leq K_{mq,Y/X} \leq K_{Y/X}$ come from Lemma 2.11, and the other ones follow by Proposition 3.13. $\square$

4. MULTIPLIER IDEAL SHEAVES

For the remainder of this paper, we work over an algebraically closed field of characteristic zero. We consider pairs of the form (X, I) , where X is a normal quasi-projective variety and $I = \sum a_k \cdot \mathcal{J}_k$ is a formal linear combination of non-zero fractional ideal sheaves on X . If each $\mathcal{J}_k$ is an ideal sheaf, $Z_k \subset X$ the subscheme defined by $\mathcal{J}_k$, and $Z = \sum a_k \cdot Z_k$ is the corresponding formal linear combination, then we identify the pairs (X, I) and (X, Z) . More generally, we allow hybrid notation by considering pairs $(X, W + J)$, where W is a formal linear combination of proper closed subschemes and J is a formal linear combination of fractional ideal sheaves (it is clear indeed that any such pair can be written in the form (X, I) for some formal linear combination I of fractional ideal sheaves).

Definition 4.1. Consider a pair (X, I) as above. A *log resolution* of (X, I) is a proper birational morphism $f: Y \rightarrow X$ from a smooth variety Y such that for every k the sheaf $\mathcal{J}_k \cdot \mathcal{O}_Y$ is the invertible sheaf of a divisor E_k on Y , the exceptional locus $\text{Ex}(f)$ of f is also a divisor, and $\text{Ex}(f) \cup E$ has simple normal crossing, where $E := \bigcup \text{Supp}(E_k)$. If Δ is a boundary on X , then a *log resolution* of the log pair (X^Δ, I) is given by a log resolution $f: Y \rightarrow X$ of (X, I) such that $\text{Ex}(f) \cup E \cup \text{Supp}(f^*(K_X + \Delta))$ has simple normal crossings.

We will need the following small generalization of Hironaka's resolution of singularities.

Theorem 4.2 ([Hir]). *Let (X, I) be a pair as above, where X is a normal quasi-projective variety defined over an algebraically closed field of characteristic zero. Then there exists a resolution of singularities of (X, I) . If Δ is a boundary on X , then there exists a resolution of singularities of (X^Δ, I) .*

Proof. Let C be a Cartier divisor on X such that $\mathcal{O}_X(C) \cdot \mathcal{J}_k \subseteq \mathcal{O}_X$ for every k , and let $W = \sum a_k \cdot W_k$, where $W_k \subset X$ is the subscheme defined by the ideal sheaf $\mathcal{O}_X(C) \cdot \mathcal{J}_k$. Then any log resolution of the pair $(X, W + \text{Supp}(C))$ (respectively, of $(X^\Delta, W + \text{Supp}(C))$) is a log resolution of (X, I) (respectively, of (X^Δ, I)). This reduces the theorem to the original version due to Hironaka. $\square$

We now restrict the attention to pairs of the form (X, Z) , where $Z = \sum a_k \cdot Z_k$ is a formal linear combination of subschemes of X . In this case, if $f: Y \rightarrow X$ is a log

resolution of X , then we denote $f^{-1}(Z) := \sum a_k \cdot f^{-1}(Z_k)$, where $f^{-1}(Z_k)$ is the divisor defined by the scheme theoretic inverse image of Z_k .

Definition 4.3. We say that Z , or (X, Z) , is *effective* if $a_k \geq 0$ for all k . If Δ is a boundary on X , then we say that the log pair (X^Δ, Z) is *effective* if so is Z .

In the following, we will always assume that the pair (X, Z) is effective (with loss of generality, one may assume for simplicity that each Z_k is an effective Cartier divisor).

Remark 4.4. With the exception of Proposition 4.13, everything in this section that is stated for effective pairs can be extended to pairs that are just *effective in codimension one*, namely, pairs $(X, \sum a_k \cdot Z_k)$ that satisfy $a_k \geq 0$ whenever $\text{codim}_X Z_k = 1$.

We start by recalling the definition of multiplier ideal sheaves for log pairs.

Definition 4.5. If Δ be a boundary on X , and consider an effective log pair (X^Δ, Z) . Let $f: Y \rightarrow X$ be a log resolution of (X^Δ, Z) . Then the *multiplier ideal sheaf* of the log pair (X^Δ, Z) is defined by

$$\mathcal{J}(X^\Delta, Z) := f_* \mathcal{O}_Y(\lceil K_{Y/X}^\Delta - f^{-1}(Z) \rceil).$$

It is well-known that $\mathcal{J}(X^\Delta, Z)$ does not depend on the choice of f and that it is a (coherent) ideal sheaf. These facts are implied by Lemma 3.12 and the following two lemmas (for more details, see [Laz]).

Lemma 4.6. *Let $f: Y \rightarrow X$ be a proper birational morphism from a smooth variety Y to a normal variety X , let P and N be effective divisors on Y without common components, and suppose that P is f -exceptional. Then $f_* \mathcal{O}_Y(P - N) = f_* \mathcal{O}_Y(-N)$.*

Proof. By a lemma of Fujita, which gives the vanishing $f_* \mathcal{O}_P(P) = 0$ (see [KMM, Lemma 1-3-2]). $\square$

Lemma 4.7. *Let $g: Y' \rightarrow Y$ be a proper birational morphism of smooth varieties, and let D be an effective $\mathbb{R}$ -divisor on Y with simple normal crossing support. Suppose that $D = S + F$, where S is a smooth reduced divisor with no common components with F and $\lfloor F \rfloor = 0$, and let S' be the proper transform of S on Y' . Then*

$$g_* \mathcal{O}_{Y'}(K_{Y'/Y} + \lceil G + S' - g^* D \rceil) = \mathcal{O}_Y$$

for every effective g -exceptional $\mathbb{R}$ -divisor G on Y' .

Proof. This is essentially due to Esnault and Viehweg [EV2]; the proof is a small generalization of [Laz, Lemma 9.2.19 and Remark 9.2.10]. $\square$

Consider now an arbitrary effective pair (X, Z) . Because of the possible failure of functoriality for composition of pullback of arbitrary $\mathbb{Q}$ -divisors (see Proposition 2.10), the definition of multiplier ideal sheaf of (X, Z) requires some preparation.

We fix a canonical divisor K_X on X . For any fixed integer $m \geq 1$, we consider a log resolution $f: Y \rightarrow X$ of the pair $(X, Z + \mathcal{O}_X(-mK_X))$, and define

$$(2) \quad \mathcal{J}_m(X, Z) := f_* \mathcal{O}_Y(\lceil K_{m,Y/X} - f^{-1}(Z) \rceil).$$

When $Z = 0$, we denote this sheaf by $\mathcal{J}_m(X)$.

The proof of the next lemma is similar to the proof of the analogous properties for multiplier ideals in the $\mathbb{Q}$ -Gorenstein case; we give it for completeness.

Lemma 4.8. *The sheaf $\mathcal{J}_m(X, Z)$ is a (coherent) sheaf of ideals on X , and its definition is independent of the choice of f .*

Proof. Let $f': Y' \rightarrow X$ be another log-resolution of $(X, Z + \mathcal{O}_X(-mK_X))$. Since we can always compare f and f' with a common resolution, we may well assume, without loss of generality, that f' factors through f and a morphism $g: Y' \rightarrow Y$. By Proposition 2.15, we have $K_{m, Y'/X} = K_{Y'/Y} + g^*K_{m, Y/X}$, and hence

$$(fg)_*\mathcal{O}_{Y'}(\lceil K_{m, Y'/X} - (fg)^{-1}(Z) \rceil) = f_*(g_*\mathcal{O}_{Y'}(K_{Y'/Y} + \lceil g^*(K_{m, Y/X} - f^{-1}(Z)) \rceil)).$$

Therefore, by Lemma 4.7, we obtain

$$(fg)_*\mathcal{O}_{Y'}(\lceil K_{m, Y'/X} - (fg)^{-1}(Z) \rceil) = f_*\mathcal{O}_Y(\lceil K_{m, Y/X} - f^{-1}(Z) \rceil).$$

This proves the independence of the definition from the choice of f . The fact that $\mathcal{J}_m(X, Z)$ is a sheaf of ideals follows from Lemma 4.6. $\square$

Lemma 4.9. *The set of ideal sheaves $\{\mathcal{J}_m(X, Z)\}_{m \geq 1}$ has a unique maximal element.*

Proof. By Proposition 3.14, we have $\mathcal{J}_m(X, Z) \subseteq \mathcal{J}_{mq}(X, Z)$ for all $m, q \geq 1$. Indeed, by Lemma 4.8, this inclusion can be verified for any choice of m and q by taking a log resolution $f: Y \rightarrow X$ of $(X, Z + \mathcal{O}_X(-mK_X) + \mathcal{O}_X(-mqK_X))$. Therefore, by Noetherian property of X , the set of ideal sheaves $\{\mathcal{J}_m(X, Z)\}_{m \geq 1}$ has a unique maximal element. $\square$

Definition 4.10. Let (X, Z) be an effective pair. The unique maximal element of $\{\mathcal{J}_m(X, Z)\}_{m \geq 1}$ is called the *multiplier ideal sheaf* of (X, Z) , and is denoted by $\mathcal{J}(X, Z)$. In the case Z is trivial, we denote the corresponding multiplier ideal sheaf by $\mathcal{J}(X)$.

Remark 4.11. Note that $\mathcal{J}(X, Z) = \mathcal{J}_m(X, Z)$ for all sufficiently divisible $m \geq 1$.

It is clear that if X is $\mathbb{Q}$ -Gorenstein, then this definition of multiplier ideal sheaf agrees with the usual one.

Proposition 4.12. *Let (X, Z) be an effective pair, and let Δ and Γ be boundaries on X with $\Delta \geq 0$ and $\Gamma \leq 0$. Then there are inclusions*

$$\mathcal{J}(X^\Delta, Z) \subseteq \mathcal{J}(X, Z) \subseteq \mathcal{J}(X^\Gamma, Z).$$

Proof. We can fix an integer $m \geq 1$ such that $m(K_X + \Delta)$ is a Cartier divisor and $\mathcal{J}(X, Z) = \mathcal{J}_m(X, Z)$. Then the proposition follows by Corollary 3.14. $\square$

As in the familiar case, multiplier ideals satisfy the following basic properties.

Proposition 4.13. *Let $Z = \sum b_k \cdot Z_k$ be an effective linear combination of proper closed subschemes of a normal variety X .*

- (a) *If $Z' = \sum b'_k \cdot Z'_k$ with $b'_k \geq b_k$ and $\mathcal{J}_{Z'_k} \subseteq \mathcal{J}_{Z_k}$ for all k , then $\mathcal{J}(X, Z') \subseteq \mathcal{J}(X, Z)$.*
- (b) *There is an $\epsilon > 0$ such that $\mathcal{J}(X, (1+t)Z) = \mathcal{J}(X, Z)$ for all $0 < t \leq \epsilon$.*

Proof. Part (a) is immediate after one fix an $m \geq 1$ such that $\mathcal{J}(X, Z) = \mathcal{J}_m(X, Z)$ and $\mathcal{J}(X, Z') = \mathcal{J}_m(X, Z')$, by comparing the respective divisors on a log resolution of $(X, Z + Z' + \mathcal{O}_X(-mK_X))$. Regarding part (b), we have $\mathcal{J}(X, (1+t')Z) \subseteq \mathcal{J}(X, (1+t)Z)$ for every $t' \geq t > 0$ by part (a), and hence there is an ϵ such that $\mathcal{J}(X, (1+t)Z)$ is the same ideal sheaf for all $0 < t \leq \epsilon$. We can then find an $m \geq 1$ and a sequence $\{t_k\}_{k \geq 1} \subset (0, \epsilon]$ converging to 0 such that $\mathcal{J}(X, (1+t_k)Z) = \mathcal{J}_m(X, (1+t_k)Z)$ and $\mathcal{J}(X, Z) = \mathcal{J}_m(X, Z)$. The same argument as in the $\mathbb{Q}$ -Gorenstein case, run on a log resolution of $(X, Z + \mathcal{O}_X(-mK_X))$, proves the assertion. $\square$

One can also extend the definition of asymptotic multiplier ideals to this context. Let D be a divisor on a normal variety X , and for every $n \geq 1$ let $B_n \subset X$ denote the base scheme of the linear system $|nD|$. We suppose that $|n_0 D| \neq \emptyset$ for some $n_0 \geq 1$, and let $N = n_0 \cdot \mathbb{Z}_+$. As in the usual case, for any given $c > 0$ the set of multiplier ideal sheaves $\{\mathcal{J}(X, \frac{c}{n} \cdot B_n)\}_{n \in N}$ has a unique maximal element (which does not depend on the choice of n_0).

Definition 4.14. The unique maximal element of $\{\mathcal{J}(X, \frac{c}{n} \cdot B_n)\}_{n \in N}$ is denoted by $\mathcal{J}(X, c \cdot \|D\|)$, and is called the *asymptotic multiplier ideal sheaf* of D with weight c .

Exactly as in the more familiar case, we have the following property (cf. Theorem 11.1.8 and Remark 11.1.13 of [Laz]).

Proposition 4.15. *With the above notation, suppose that $\mathcal{J}(X) = \mathcal{O}_X$ (i.e., X is ‘log terminal’, see Definition 8.1 below). Then $\mathcal{J}(X, \|D\|)$ contains the ideal sheaf of the base scheme of $|D|$. In particular, we obtain*

$$H^0(\mathcal{O}_X(D) \cdot \mathcal{J}(X, \|D\|)) \cong H^0(\mathcal{O}_X(D)).$$

More generally, we can define a combination of multiplier ideals and asymptotic multiplier ideals. This construction will be used to study adjunction properties in Section 6.

Let (X, Z) be an effective pair, and let $\mathbf{a}_\bullet = (\mathbf{a}_m)_{m \geq 0}$ be a sequence of non-zero ideal sheaves $\mathbf{a}_m \subseteq \mathcal{O}_X$. We do not require that $\mathbf{a}_\bullet$ be a graded system of ideals. Instead, we only assume that the sequence of fractional ideal sheaves $(\mathbf{a}_m \cdot \mathcal{O}_X(-mK_X))_{m \geq 0}$ be a graded system. Notice that this is a weaker condition (the two conditions are equivalent if K_X is Cartier, and if X is $\mathbb{Q}$ -Gorenstein then they are equivalent up to some truncation of the systems).

Lemma 4.16. *The set of ideal sheaves $\{\mathcal{J}_m(X, Z + \frac{1}{m} \cdot V(\mathbf{a}_m))\}_{m \geq 0}$ has a unique maximal element.*

Proof. For any $m, q \geq 1$, let $f: Y \rightarrow X$ be a log resolution of $(X, \mathcal{O}_X(mK_X) + \mathcal{O}_X(mqK_X) + \mathbf{a}_m + \mathbf{a}_{mq})$, and write $\mathbf{a}_m \cdot \mathcal{O}_Y = \mathcal{O}_Y(-A_m)$ and $\mathbf{a}_{mq} \cdot \mathcal{O}_Y = \mathcal{O}_Y(-A_{mq})$. We have $K_{m,Y/X} - \frac{1}{m}A_m \leq K_{mq,Y/X} - \frac{1}{mq}A_{mq}$, and hence $\mathcal{J}_m(X, Z + \frac{1}{m} \cdot V(\mathbf{a}_m)) \subseteq \mathcal{J}_{mq}(X, Z + \frac{1}{mq} \cdot V(\mathbf{a}_{mq}))$. The lemma follows. $\square$

Definition 4.17. We denoted the maximal element $\{\mathcal{J}_m(X, Z + \frac{1}{m} \cdot V(\mathbf{a}_m))\}_{m \geq 0}$ by $\mathcal{J}(X, Z + \|\mathbf{a}_\bullet\|)$, and call it the *asymptotic multiplier ideal* of $(X, Z + \mathbf{a}_\bullet)$.

Remark 4.18. If $\mathbf{a}_m = 0$ for all sufficiently divisible $m \geq 1$, then $\mathcal{J}(X, Z + \|\mathbf{a}_\bullet\|) = \mathcal{J}(X, Z)$.

Remark 4.19. Hara’s generalization of $\tau(X, Z)$ on toric varieties... ²

5. VANISHING THEOREM AND FIRST APPLICATIONS

The main result of this section is the following extension of Nadel’s vanishing theorem ([EV1, Nad1, Nad2]; see also [Laz, Section 9.4])

Theorem 5.1. *Consider a pair (X, Z) , where X is a projective normal variety defined over an algebraically closed field of characteristics zero and $Z = \sum_{k=1}^n a_k \cdot Z_k$ is an effective formal linear combination of proper closed subschemes of X , and let Δ be a boundary on*

²Ask Hara about his generalization.

X of the form $\Delta = -D$, where D is an effective divisor. Let $m \geq 1$ be an integer such that $\mathcal{J}(X, Z) = \mathcal{J}_m(X, Z)$. Then the following property holds: if A and B_k , for $1 \leq k \leq n$, are Cartier divisors for which the sheaves $\mathcal{O}_X(A) \otimes \mathcal{O}_X(-mD)$ and $\mathcal{O}_X(B_k) \otimes \mathcal{J}_{Z_k}$ for $1 \leq k \leq n$ are globally generated, and L is a Cartier divisor such that

$$L - (K_X + \Delta + \frac{1}{m}A + \sum a_k B_k)$$

is nef and big, then

$$H^i(\mathcal{O}_X(L) \otimes \mathcal{J}(X, Z)) = 0 \quad \text{for } i > 0.$$

Proof. Let $m \geq 1$ be a sufficiently divisible integer so that $\mathcal{J}(X, Z) = \mathcal{J}_m(X, Z)$. If $f: Y \rightarrow X$ is a log resolution of $(X, Z + \mathcal{O}_X(-mK_X) + \mathcal{O}_X(-mD))$ and E is the divisor on Y such that $\mathcal{O}_X(-mD) \cdot \mathcal{O}_Y = \mathcal{O}_Y(-E)$, then we have

$$\mathcal{J}(X, Z) = \mathcal{J}_m(X, Z) = f_*\mathcal{O}_Y([K_Y - f^*(K_X + \Delta) - \frac{1}{m}E - f^{-1}(Z)]).$$

By the hypothesis in the theorem, we can find a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor M on X that is $\mathbb{Q}$ -linearly equivalent to $\frac{1}{m}A + \sum b_k B_k$ and such that

$$[K_Y - f^*(K_X + \Delta) - \frac{1}{m}E - f^{-1}(Z)] = [K_Y - f^*(K_X + \Delta) - f^*M].$$

Then

$$\mathcal{O}_X(L) \otimes \mathcal{J}(X, Z) = f_*\mathcal{O}_Y(K_Y + [f^*(L - (K_X + \Delta + M))]).$$

Since $L - (K_X + \Delta + M)$ is nef and big, the Kawamata–Viehweg vanishing theorem (see [Laz, Theorem 9.1.18]) yields

$$H^i(\mathcal{O}_Y(K_Y + [f^*(L - (K_X + \Delta + M))])) = 0 \quad \text{for } i > 0.$$

On the other hand, the local vanishing theorem (see [Laz, Theorem 9.4.1 and Generalization 9.1.22]) gives

$$R^i f_*\mathcal{O}_Y(K_Y + [f^*(L - (K_X + \Delta + M))])) = 0 \quad \text{for } i > 0$$

(this vanishing also holds only assuming that X is quasi-projective). Then the theorem follows via the Leray spectral sequence. $\square$

As a special case of the theorem and its proof, we obtain the following immediate corollary.

Corollary 5.2. *Let X be a normal quasi-projective variety defined over an algebraically closed field of characteristic zero and $m > 0$ an integer such that $\mathcal{J}(X) = \mathcal{J}_m(X)$. Then for any log resolution $f: Y \rightarrow X$ of the pair $(X, \mathcal{J}_{mK_X})$ we have*

$$R^i f_*\mathcal{O}_Y([K_{m,Y/X}]) = 0 \quad \text{for } i > 0.$$

Arguing as in of Section 9.4.E of [Laz], we also have the following application.

Corollary 5.3. *With the same notation and assumptions as in Theorem 5.1, let H be a very ample Cartier divisor on X . Then the sheaf*

$$\mathcal{O}_X(L + kH) \otimes \mathcal{J}(X, Z)$$

is globally generated for every integer $k \geq \dim X + 1$.

The following lemma is a trivial consequence of Fujita's lemma and the projection formula when the divisor G is Cartier.

Lemma 5.4. *Let G be a divisor on a normal variety X , and suppose that $f: Y \rightarrow X$ is a proper birational morphism from a smooth variety such that $\mathcal{O}_Y \cdot \mathcal{O}_X(G)$ is an invertible sheaf. Suppose that N and P are effective divisors on Y with no common components and such that P is f -exceptional. Then $f_*(\mathcal{O}_Y(P - N) \cdot \mathcal{O}_X(G)) = \mathcal{O}_X(G)$ if and only if $N = 0$.*

Proof. Suppose that $N = 0$. Then we have the homomorphism

$$\mathcal{O}_X(G) = f_*(\mathcal{O}_Y \cdot \mathcal{O}_X(G)) \rightarrow f_*(\mathcal{O}_Y(P) \cdot \mathcal{O}_X(G)) \rightarrow \mathcal{O}_X(G)$$

where the last map is given by the fact that as P is exceptional, $\mathcal{O}_X(G) = f_*(\mathcal{O}_Y(P) \cdot \mathcal{O}_X(G))^{**}$. It is now clear that $f_*(\mathcal{O}_Y(P) \cdot \mathcal{O}_X(G)) = \mathcal{O}_X(G)$. On the other hand, if $N \neq 0$, then by Lemma 4.6 $f_*\mathcal{O}_Y(P - N) = f_*\mathcal{O}_Y(-N) \neq \mathcal{O}_X$. If $f_*(\mathcal{O}_Y(P - N) \cdot \mathcal{O}_X(G)) = \mathcal{O}_X(G)$ then there is a homomorphism

$$\mathcal{O}_Y \cdot \mathcal{O}_X(G) = \mathcal{O}_Y \cdot f_*(\mathcal{O}_Y(P - N) \cdot \mathcal{O}_X(G)) \rightarrow \mathcal{O}_Y(P - N) \cdot \mathcal{O}_X(G)$$

so that $h^0(\mathcal{O}_Y(P - N)) = h^0(\mathcal{O}_Y(-N)) > 0$, and thus $N = 0$. $\square$

We close the section with a generalization of Shokurov–Kollár’s connectedness lemma [Kol1, Sho].

Corollary 5.5. *With the same notation and assumptions as in Theorem 5.1, let $K_{m,Y/X} - f^{-1}(Z) = \sum e_i E_i = A - B$ where $A = \sum_{e_i > -1} e_i E_i$. Assume that $[A]$ is exceptional, i.e., that all divisorial components of Z appear with coefficient less than 1. Then $\text{Supp}(B)$ is connected in a neighborhood of any fiber of f . If moreover B is irreducible and reduced, then $f(B)$ is normal.*

Proof. Since $R^1 f_* \mathcal{O}_Y([A - B]) = 0$, there is a short exact sequence

$$0 \rightarrow f_* \mathcal{O}_Y([A - B]) \rightarrow f_* \mathcal{O}_Y([A]) \rightarrow f_* \mathcal{O}_{[B]}([A]) \rightarrow 0.$$

As $[A]$ is exceptional, $f_* \mathcal{O}_Y([A]) = \mathcal{O}_X$ by Lemma 5.4, and both assertions follow easily (cf. [Kol2, Theorem 7.4] and [Kaw4, Theorem 1.6]). $\square$

6. ADJOINT IDEAL SHEAVES

This section is devoted to a discussion on adjoint ideal sheaves. As in the previous section, we work over an algebraically closed field of characteristic zero. The usual definition appearing in the literature can be generalized to a more general context in the following way. Let (X, Z) be an effective pair as in the previous section, and let H be a $\mathbb{Q}$ -divisor on X . We assume that $H = \sum h_i H_i$, where H_i are distinct prime divisors not contained in the support of Z and $h_i \leq 1$. For any positive integer m such that mH is a divisor, we fix a log resolution $f: Y \rightarrow X$ of $(X, Z + \mathcal{O}_X(-mK_X) + \mathcal{O}_X(-mH))$ such that all components of the proper transform H_Y of H on Y are disconnected; if Δ is a given boundary on X , then we also suppose that f is a log resolution of the log pair $(X^\Delta, Z + \mathcal{O}_X(-mH))$. Then we consider the ideal sheaves

$$\text{adj}_{m,H}(X, Z) := f_* \mathcal{O}_Y([K_{m,Y/X} - f^{-1}(Z) - \frac{1}{m} \cdot f^\flat(mH) + H_Y])$$

and

$$\text{adj}_{m,H}(X^\Delta, Z) := f_* \mathcal{O}_Y([K_{Y/X}^\Delta - f^{-1}(Z) - \frac{1}{m} \cdot f^\flat(mH) + H_Y]).$$

Similar proofs of those of Lemmas 4.8 and 4.9 give the following analogous statements.

Lemma 6.1. *The sheaves $\text{adj}_{m,H}(X, Z)$ and $\text{adj}_{m,H}(X^\Delta, Z)$ are (coherent) sheaves of ideals on X , and their definitions are independent of the choice of f .*

Proof. The proof is similar to that of Lemma 4.8. $\square$

Lemma 6.2. *Both sets of ideal sheaves $\{\text{adj}_{m,H}(X, Z)\}_{m \geq 1}$ and $\{\text{adj}_{m,H}(X^\Delta, Z)\}_{m \geq 1}$ have a unique maximal element.*

Proof. This follows by the same argument as in the proof of Lemma 4.9. $\square$

Definition 6.3. The maximal element of $\{\text{adj}_{m,H}(X, Z)\}_{m \geq 1}$ is called the *adjoint ideal sheaf* of the pair (X, Z) along H , and is denoted by $\text{adj}_H(X, Z)$. Similarly, the maximal element of $\{\text{adj}_{m,H}(X^\Delta, Z)\}_{m \geq 1}$ is called the *adjoint ideal sheaf* of the log pair (X^Δ, Z) along H , and is denoted by $\text{adj}_H(X^\Delta, Z)$.

Remark 6.4. As for multiplier ideal sheaves, we have $\text{adj}_H(X, Z) = \text{adj}_{m,H}(X, Z)$ and $\text{adj}_H(X^\Delta, Z) = \text{adj}_{m,H}(X^\Delta, Z)$ for all sufficiently divisible $m \geq 1$.

The terminology for this ideal sheaf comes from the particular setting in which H is an effective normal Cartier divisor. In this case, observing that $X \setminus X_{\text{reg}}$ has codimension ≥ 2 in X , and similarly $H \setminus X_{\text{reg}}$ has codimension ≥ 2 in H , one sees that for every divisor D on X with no common components with H , its restriction $D|_H$ to H is well defined as a divisor on H . It is then convenient to fix a canonical divisor K_X on X such that $K_X + H$ had order zero along the components of H . Since H is a normal divisor on X that is Cartier in codimension 2, we can apply adjunction in codimension 1 (both on X and on H) to conclude that $K_H := (K_X + H)|_H$ is a canonical divisor on H (cf. Remark 5.47 in [KM]). However, $\mathcal{O}_X(-m(K_X + H)) \cdot \mathcal{O}_H$ may in general differ from $\mathcal{O}_H(-mK_H)$ if $K_X + H$ is not $\mathbb{Q}$ -Cartier.

Definition 6.5. With the above notation, for every $m \geq 0$ we define the *m-th adjunction defect ideal* of H in X to be the ideal sheaf $\mathfrak{a}_{m,H/X} \subseteq \mathcal{O}_H$ determined by the formula

$$\mathcal{O}_X(-m(K_X + H)) \cdot \mathcal{O}_H = \mathfrak{a}_{m,H/X} \cdot \mathcal{O}_H(-mK_H).$$

We call $\mathfrak{a}_{\bullet,H/X} = (\mathfrak{a}_{m,H/X})_{m \geq 0}$ the *adjunction defect system* of H in X .

It is easy to see that this definition is independent of the choice of K_X . Note that $\mathfrak{a}_{m,H/X}$ is cosupported on a set of codimension ≥ 2 in H . Moreover, if X is $\mathbb{Q}$ -Gorenstein, then $K_X + H$ is $\mathbb{Q}$ -Cartier and hence $\mathfrak{a}_{m,H/X} = \mathcal{O}_H$ for all sufficiently divisible m .

Lemma 6.6. *The sequence of ideals $(\mathfrak{a}_{m,H/X} \cdot \mathcal{O}_X(-mK_H))_{m \geq 0}$ forms a graded system.*

Proof. It follows by taking the image in $\mathcal{O}_H$ of the inclusion $\mathcal{O}_X(-m(K_X + H)) \cdot \mathcal{O}_X(-n(K_X + H)) \subseteq \mathcal{O}_X(-(m+n)(K_X + H))$. $\square$

Theorem 6.7. *Suppose that H is a normal Cartier divisor on X with no components contained in the support of Z . Then the adjoint ideal $\text{adj}_H(X, Z)$ sits in the exact sequence*

$$0 \rightarrow \mathcal{J}(X, Z) \otimes \mathcal{O}_X(-H) \rightarrow \text{adj}_H(X, Z) \rightarrow \mathcal{J}(H, Z|_H + \|\mathfrak{a}_{\bullet,H/X}\|) \rightarrow 0,$$

where $\mathfrak{a}_{\bullet,H/X}$ is the adjunction defect system of H in X .

Proof. Let $m > 0$ be a sufficiently divisible integer, so that $\mathcal{J}(X, Z) = \mathcal{J}_m(X, Z)$ and $\mathcal{J}(H, Z|_H + \|\mathfrak{a}_{\bullet,H/X}\|) = \mathcal{J}_m(H, Z|_H + \frac{1}{m} \cdot V(\mathfrak{a}_{m,H/X}))$, and let $f: Y \rightarrow X$ be a log resolution of $(X, Z + \mathcal{J}_{mK_X} + \mathcal{O}_X(-mH) + \mathfrak{a}_{m,H/X})$ such that all components of the proper transform

H_Y of H are disjoint. Denote $h: H_Y \rightarrow H$ the restriction of f . Notice that $\frac{1}{m} \cdot f^{\natural}(mH) = f^*H$. We have the short exact sequence

$$\begin{aligned} 0 \rightarrow \mathcal{O}_Y([K_{m,Y/X} - f^{-1}(Z) - f^*H]) &\rightarrow \mathcal{O}_Y([K_{m,Y/X} - f^{-1}(Z) - f^*H] + H_Y) \\ &\rightarrow \mathcal{O}_{H_Y}([K_{m,Y/X} - f^{-1}(Z) - f^*H + H_Y]) \rightarrow 0. \end{aligned}$$

By Theorem 5.1 and its proof, we have that $R^1 f_* \mathcal{O}_Y([K_{m,Y/X} - f^{-1}(Z) - f^*H]) = 0$. Therefore, it suffices to show that

$$(3) \quad (K_{m,Y/X} - f^*H + H_Y)|_{H_Y} = K_{m,H_Y/H} - \frac{1}{m} \cdot f^{-1}V(\mathfrak{a}_{m,H/X})$$

as $\mathbb{Q}$ -divisors. We fix a canonical divisor K_X on X such that $K_X + H$ had order zero along the components of H , and as usual let K_Y be such that $f_* K_Y = K_X$. Note that the coefficients of the components of H_Y in $K_Y + H_Y$ are also zero. By adjunction, $K_H := (K_X + H)|_H$ is a canonical divisor on H and $K_{H_Y} := (K_Y + H_Y)|_{H_Y}$ is a canonical divisor on H_Y . Note that $h_* K_{H_Y} = K_H$. We have $\mathcal{O}_X(-m(K_X + H)) \cdot \mathcal{O}_H = \mathfrak{a}_{m,H/X} \cdot \mathcal{O}_H(-mK_H)$, and this implies that

$$f^{\natural}(m(K_X + H))|_{H_Y} = h^{\natural}(mK_H) + f^{-1}V(\mathfrak{a}_{m,H/X}).$$

The equality (3) follows. $\square$

We immediately obtain the following inversion of adjunction statement.

Corollary 6.8. *In the hypothesis of Theorem 6.7, we have $\text{adj}_H(X, Z) = \mathcal{O}_X$ in a neighborhood of H if and only if $\mathcal{J}(H, Z|_H + \|\mathfrak{a}_{\bullet, H/X}\|) = \mathcal{O}_H$. In particular, we have $\mathcal{J}(H, Z|_H) = \mathcal{O}_H$ whenever $\text{adj}_H(X, Z) = \mathcal{O}_X$ in a neighborhood of H , and the two conditions are equivalent if $\mathfrak{a}_{m,H/X} = \mathcal{O}_H$ for all sufficiently divisible $m \geq 1$.*

Remark 6.9. The corollary above should be compared with the following well known statement: If $S \subset X$ is a normal Cartier divisor (in fact, it suffices to assume that S is Cartier in codimension 2) on a normal variety and B is an effective divisor such that $K_X + S + B$ is $\mathbb{Q}$ -Cartier, then $(X, S + B)$ is purely log terminal with respect to S if and only if $(S, B|_S)$ is Kawamata log terminal (see for example [KM, Theorem 5.50]).

In the special case when $\mathfrak{a}_{m,H/X} = \mathcal{O}_H$ for all sufficiently divisible m , we also obtain the following vanishing theorem for adjoint ideals.

Theorem 6.10. *With the same assumptions as in Theorem 5.1, let H be a normal Cartier divisor on X , and suppose that $\mathfrak{a}_{m,H/X} = \mathcal{O}_H$ for all sufficiently divisible $m \geq 1$. Then*

$$H^i(\mathcal{O}_X(L + H) \otimes \text{adj}_H(X, Z)) = 0 \quad \text{for } i > 0.$$

Proof. We have

$$H^i(\mathcal{O}_X(L) \otimes \mathcal{J}(X, Z)) = 0 \quad \text{for } i > 0$$

by Theorem 5.1. Observe that the restriction $\Delta|_H$ of Δ to H is a boundary on H , and the sheaves $\mathcal{O}_H(A) \otimes \mathcal{O}_H(-mD|_H)$ and $\mathcal{O}_H(B_k) \otimes \mathcal{J}_{Z_k|_H}$ are globally generated. Moreover, $(L - (K_X + \Delta + \frac{1}{m}A + \sum a_k B_k))|_H$ is nef and big, and hence

$$(L + H)|_H - (K_H + \Delta|_H + \frac{1}{m}A|_H + \sum a_k B_k|_H)$$

is nef and big. Then the same theorem, applied to $(H, Z|_H)$, gives the vanishings

$$H^i(\mathcal{O}_H(L + H) \otimes \mathcal{J}(H, Z|_H)) = 0 \quad \text{for } i > 0.$$

The assertion then follows by Theorem 6.7. $\square$

We close this section by observing that in general multiplier ideals can be realized as suitable adjoint ideals of log pairs.

Proposition 6.11. *Let (X, Z) be an effective pair, and let Δ be a boundary on X of the form $\Delta = \sum d_i \Delta_i$, where Δ_i are prime divisors not contained in the support of Z and $d_i \geq -1$. Then*

$$\mathcal{J}(X, Z) = \text{adj}_{-\Delta}(X^\Delta, Z).$$

Proof. We can fix $m \geq 1$ such that $m(K_X + \Delta)$ is a Cartier divisor and, moreover, $\mathcal{J}(X, Z) = \mathcal{J}_m(X, Z)$ and $\text{adj}_{-\Delta}(X^\Delta, Z) = \text{adj}_{m, -\Delta}(X^\Delta, Z)$. Then the assertion follows by Proposition 3.13. $\square$

7. SINGULARITIES OF LOG PAIRS

Let X be a normal variety defined over an algebraically closed field of characteristic zero, and consider an effective formal linear combination $Z = \sum a_k \cdot Z_k$ of proper closed subschemes $Z_k \subset X$. Let $\text{Sing}(X)$ denote the singular locus of X . Supposing that Δ is a boundary on X (e.g., $\Delta = 0$ if X is $\mathbb{Q}$ -Gorenstein), several definitions of singularities have been given (and extensively used) in the literature for the log pair (X^Δ, Z) . These notions, which we recall next, have played a fundamental part in the study of the birational geometry of higher dimensional varieties.

To simplify the terminology, we say that F is prime divisor *over* X if F is an irreducible and reduced divisor on a normal variety X' with a given birational morphism $\mu: X' \rightarrow X$. We also say that F is *exceptional* over X if its center $c_X(F)$ has codimension ≥ 2 .

Consider an effective log pair (X^Δ, Z) . We define the *log discrepancy* of a prime divisor F over X with respect to (X^Δ, Z) to be

$$a_F(X^\Delta, Z) := \text{ord}_F(K_{X'/X}^\Delta) + 1 - \text{val}_F(Z).$$

Definition 7.1. The log pair (X^Δ, Z) is said to be *log canonical* (resp., *log terminal*) if $a_F(X^\Delta, Z) \geq 0$ (resp., > 0) for every prime divisor F over X , and it is said to be *canonical* (resp., *terminal*) if $a_F(X^\Delta, Z) \geq 1$ (resp., > 1) for every exceptional prime divisor F over X . We say that the log variety X^Δ is *log canonical* (resp., *log terminal*, *canonical*, *terminal*) if so is the log pair $(X^\Delta, 0)$.

Remark 7.2. Suppose that Δ and Z do not share any (codimension one) irreducible component. Then, comparing the above definition of log terminal singularities for a log pair (X^Δ, Z) with the analogous definitions given in [KM], our definition can be interpreted by saying that the log terminal singularities of (X^Δ, Z) are ‘pure’ with respect to Δ and ‘Kawamata’ with respect to Z (cf. Remark 3.11). If one adds the condition that $\lfloor \Delta \rfloor \leq 0$ then the above definition reduces to the usual one of Kawamata log terminal singularities.

Definition 7.3. Let (X^Δ, Z) be an effective pair, and assume that X^Δ is log terminal. The *log canonical threshold* is defined by

$$\text{lc}(X^\Delta, Z) := \sup\{t > 0 \mid (X^\Delta, tZ) \text{ is log terminal}\}.$$

The following property is well known.

Proposition 7.4. *An effective log pair (X^Δ, Z) is log terminal if and only if $\mathcal{J}(X^\Delta, Z) = \mathcal{O}_X$. Moreover, if X^Δ is log terminal, then*

$$\text{lc}(X^\Delta, Z) = \sup\{t > 0 \mid \mathcal{J}(X^\Delta, tZ) = \mathcal{O}_X\}.$$

8. LOG TERMINAL AND LOG CANONICAL SINGULARITIES

In this section we extend the definitions of log terminal and log canonical singularities of pairs to our more general context, and discuss some generalizations to this context of certain results on rational and log terminal singularities due, respectively, to Elkik and Kawamata. The ground field is still algebraically closed and of characteristic zero.

Let (X, Z) be an effective pair. For any integer $m \geq 1$, we define the *m-th limiting log discrepancy* of (X, Z) to be

$$a_{m,F}(X, Z) := \text{ord}_F(K_{m,X'/X}) + 1 - \text{val}_F(Z).$$

Definition 8.1. The pair (X, Z) is said to be *log canonical* (resp., *log terminal*) if there is an integer m_0 such that $a_{m,F}(X, Z) \geq 0$ (resp., > 0) for every prime divisor F over X and any positive multiple m of m_0 . If X is log terminal, then the *log canonical threshold* of (X, Z) is

$$\text{lc}(X, Z) := \sup\{t > 0 \mid (X, tZ) \text{ is log terminal}\}.$$

Clearly these notions coincide with the usual ones when X is $\mathbb{Q}$ -Gorenstein, and in general, if (X, Z) is log terminal, then it is log canonical. One can also define the *jumping numbers* of an effective pair (X, Z) analogously to the familiar case.

Proposition 8.2. *An effective pair (X, Z) is log terminal if and only if $\mathcal{J}(X, Z) = \mathcal{O}_X$. Moreover, if X is log terminal, then*

$$\text{lc}(X, Z) = \sup\{t > 0 \mid \mathcal{J}(X, tZ) = \mathcal{O}_X\}.$$

Proof. Let q be a positive integer such that $\mathcal{J}(X, Z) = \mathcal{J}_q(X, Z)$. If (X, Z) is log terminal and m_0 is as in the definition, then $\mathcal{J}(X, Z) = \mathcal{J}_{qm_0}(X, Z) = \mathcal{O}_X$. Conversely, we just take $m_0 = q$. $\square$

Proposition 8.3. *Let Δ and Γ be boundaries on X with $\Delta \geq 0$ and $\Gamma \leq 0$, and consider the following three assertions:*

- (a) *The log pair (X^Δ, Z) is log terminal (resp., log canonical).*
- (b) *The pair (X, Z) is log terminal (resp., log canonical).*
- (c) *The log pair (X^Γ, Z) is log terminal (resp., log canonical).*

Then (a) $\Rightarrow$ (b) $\Rightarrow$ (c).

Proof. By Proposition 4.12. $\square$

Corollary 8.4. *With the assumptions as in Proposition 8.3, suppose that X^Δ is log terminal. Then*

$$\text{lc}(X^\Delta, Z) \leq \text{lc}(X, Z) \leq \text{lc}(X^\Gamma, Z).$$

We next address the extension of Elkik's theorem on rational singularities [Elk].

Theorem 8.5. *Let X be a normal variety with log terminal singularities. Then X has rational singularities.*

Proof. The proof follows from Corollary 5.2 and the arguments of Theorem 5.22 of [KM]. $\square$

Question 8.6. Let X be a normal Cohen–Macaulay variety with log canonical singularities. Is it true then that X has Du Bois singularities? ³

³Kovacs–Schweede–Smith should have proven for Cohen–Macaulay varieties that Du Bois singularities are characterized by the equation $f_*(\omega_Y \otimes \mathcal{O}_Y(\text{Ex}(f))) = \omega_X$, and used this to prove that a normal Cohen–Macaulay variety with log canonical singularities has Du Bois singularities.

In [Kaw1], Kawamata proves an important result on the singularities of minimal log canonical centers, which in particular implies that such centers have rational singularities. Using Proposition 8.3, Kawamata's theorem immediately implies the following statement.

Theorem 8.7. *In the setting and terminology of Theorem 1 in [Kaw1], ‘minimal log canonical centers’ are normal varieties with log terminal singularities (in the sense introduced in this paper).*

Kawamata's subadjunction theorem appears to be a natural setting for the theory developed in this paper, as in general ‘minimal log canonical centers’ are not known to be $\mathbb{Q}$ -Gorenstein (even when the ambient variety is smooth). In fact, it turns out that, under some additional conditions, the assumptions in the above result can be extended to our general setting.

Suppose that (X, Z) is an effective pair that is *strictly log canonical*, namely, that is log canonical but not log terminal. Let m_0 as in Definition 8.1.

Definition 8.8. A subvariety $W \subset X$ is said to be a *log canonical center* of (X, Z) if there is an exceptional prime divisor E over X such that $c_X(E) = W$ and $a_{m,E}(X, Z) = 0$ for some (equivalently, for every) multiple m of m_0 . A log canonical center is said to be *minimal* if it is so with respect to inclusions. A log canonical center $W \subset X$ is said to be a *critical variety* for (X, Z) if there is a unique E with $c_X(E) \cap W \neq \emptyset$ and $a_{m,E}(X, Z) = 0$ for some (equivalently, for every) multiple m of m_0 .

Remark 8.9. If W is a critical variety for (X, Z) , then there is a unique E with $c_X(E) = W$ and $a_{m,E}(X, Z) = 0$ for any multiple m of m_0 , and the valuation associated to the divisor E is independent of m .

Theorem 8.10. *Every critical variety $W \subset X$ for an effective strictly log canonical pair (X, Z) is a normal variety with log terminal singularities.*

Proof. The arguments are an adaptation of those in [Kaw1]. Note that W is normal by Corollary 5.5. We fix a boundary Δ on X such that $-\Delta$ is an effective divisor, fix a sufficiently divisible m , and let $\mu: Y \rightarrow X$ be a log resolution of $(X, Z + \mathcal{O}_X(-mK_X) + \mathcal{O}_X(m\Delta))$. Then we have

$$K_{m,Y/X} - \mu^{-1}(Z) = -E - F,$$

where E is a prime divisor mapping to W and F is a $\mathbb{Q}$ -divisor whose support does not contain E and whose coefficients in a neighborhood of $\mu^{-1}(W)$ (and hence also of E) are less than one. In particular, we have $\lceil -F|_E \rceil \geq 0$, since $E + F$ has simple normal crossing support.

Denote by $\nu: E \rightarrow W$ the restriction of μ . As in the proof of Theorem 5.1, we have $R^1\mu_*\mathcal{O}_Y(-E + \lceil -F \rceil) = 0$, and thus the natural homomorphism $\mu_*\mathcal{O}_Y(\lceil -F \rceil) \rightarrow \nu_*\mathcal{O}_E(\lceil -F|_E \rceil)$ is surjective. Since the components of F with negative coefficients are μ -exceptional, we conclude that the natural homomorphism $\mathcal{O}_W \rightarrow \nu_*\mathcal{O}_E(\lceil -F|_E \rceil)$ is an isomorphism.

Arguing as in the proof of Theorem 5.1, we can find a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor M on X such that

$$K_{m,Y/X} - \mu^{-1}(Z) = K_Y - \mu^*(K_X + \Delta) - \mu^*M + \Sigma,$$

where Σ is a divisor with no common components with E or F , and such that $\lfloor \Sigma \rfloor = 0$. We can also assume that $E + F + \Sigma$ has simple normal crossing support, so that, in particular,

$\Sigma|_E$ has no common components with $F|_E$ and $F|_E + \Sigma|_E$ has simple normal crossing support. Denoting for short $F' := F + \Sigma$, we have

$$K_E + F'|_E = \mu^*(K_X + \Delta + M)|_E.$$

Notice that $\lceil -F'|_E \rceil = \lceil -F|_E \rceil - \lfloor \Sigma \rfloor = \lceil -F|_E \rceil$.

We assume that ν factors through a resolution of singularities $\sigma: V \rightarrow W$ and a morphism $f: E \rightarrow V$. We have

$$K_E + F'|_E = (K_Y + E + F')|_E \sim_{\mathbb{Q}} \mu^*(K_X + \Delta + M)|_E.$$

Since $K_X + \Delta + M$ is a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor, after possibly replacing Y and V by appropriate birational models, we are in the setting of Theorem 2 of [Kaw1], which implies that

$$K_E + F'|_E \sim_{\mathbb{Q}} f^*(K_V + \Delta_V + M_V),$$

here M_V is a nef $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor, Δ_V is an effective boundary on V , and V^{Δ_V} is a log variety with log terminal singularities. We may pick a general hyperplane section H of X and an effective $\mathbb{Q}$ -divisor $\Gamma \sim_{\mathbb{Q}} M_V + \sigma^*(H|_W)$ such that $V^{\Delta_V + \Gamma}$ is still log terminal. We fix the canonical divisor $K_W := \sigma_* K_V$ on W , and let $\Delta_W := \sigma_*(\Delta_V + \Gamma)$. Then we have

$$K_W + \Delta_W \sim_{\mathbb{Q}} (K_X + \Delta + M + H)|_W,$$

which in particular implies that $K_W + \Delta_W$ is $\mathbb{Q}$ -Cartier, and

$$\sigma^*(K_W + \Delta_W) \sim_{\mathbb{Q}} K_V + \Delta_V + \Gamma.$$

By the Negativity Lemma ([KM, Lemma 3.39]), this last formula is an equality. We deduce that the log variety W^{Δ_W} has log terminal singularities. Since Δ_W is effective, it follows by Proposition 8.3 that W has log terminal singularities. $\square$

We close this section with a discussion on surface singularities. As explained in [KM, Notation 4.1], one can define the notions of *numerically log terminal* and *numerically log canonical* singularities for arbitrary normal surfaces, using the perfect pairing on the relative Néron–Severi space of a resolution. Here we show that a normal surface is log terminal (resp., log canonical) if and only if it is numerically log terminal (resp., numerically log canonical).

Proposition 8.11. *Let X be a normal surface with log terminal singularities. Then X is $\mathbb{Q}$ -factorial.*

Proof. We can assume that X is projective. By Theorem 8.5, X has rational singularities. Let $f: Y \rightarrow X$ be a log resolution, and consider the commutative diagram

$$\begin{array}{ccccc} H^1(\mathcal{O}_Y) & \longrightarrow & \text{Pic}(Y) & \twoheadrightarrow & \text{NS}(Y) \\ \uparrow \cong & & \uparrow & & \uparrow \\ H^1(\mathcal{O}_X) & \longrightarrow & \text{Pic}(X) & \twoheadrightarrow & \text{NS}(X) \end{array}$$

where the vertical arrows are given by pulling back via f . The first one of these is an isomorphism because X has rational singularities.

Fix an arbitrary divisor D on X and let D_Y be its strict transform on Y . Let E_i be the prime exceptional divisors of f . Since the matrix $(E_i \cdot E_j)$ is negative definite, we may find an exceptional $\mathbb{Q}$ -divisor $E = \sum e_i E_i$ such that $D_Y + E \equiv_f 0$. After fixing an integer $m > 0$ such that mE is integral, this implies that the image $[m(D_Y + E)]$ in $\text{NS}(Y)$ is the pull back of a class $[B] \in \text{NS}(X)$ of some Cartier divisor B on X . Note that

$[m(D_Y + E) - f^*B] = 0$ in $\text{NS}(Y)$. By the exactness (in the middle) of the top row, there is an element $\alpha \in H^1(\mathcal{O}_Y)$ mapping to $m(D_Y + E) - f^*B$. Then the image $A \in \text{Pic}(X)$ of the element $(f^*)^{-1}(\alpha) \in H^1(\mathcal{O}_X)$ is a Cartier divisor pulling back to $m(D_Y + E) - f^*B$. This implies that $m(D_Y + E)$ is the pull-back of the Cartier divisor $A + B$ on X , and so as $mD = f_*(mD_Y) = A + B$, the divisor D is $\mathbb{Q}$ -Cartier. $\square$

Corollary 8.12. *A normal surface X is log terminal if and only if it is numerically log terminal.*

Proof. Immediate since by Proposition 8.11 and by Proposition 4.11 of [KM], if X is log terminal or is numerically log terminal, then X is $\mathbb{Q}$ -factorial. $\square$

Proposition 8.13. *A normal surface X is log canonical if and only if it is numerically log canonical.*

Proof. If X is numerically log canonical, then it is $\mathbb{Q}$ -Gorenstein (cf. [KM, Notation 4.1]), and hence log canonical. Conversely, suppose that X is log canonical in the generality introduced in this section. Let K_Y be a canonical divisor on Y , and let $K_X = f_*K_Y$. We fix a sufficiently divisible $m \geq 1$. Let $f: Y \rightarrow X$ be a log resolution of $(X, \mathcal{O}_X(-mK_X))$, and write $\mathcal{O}_X(-mK_X) \cdot \mathcal{O}_Y = \mathcal{O}_Y(-A)$ where A is a divisor on Y . Note that $-A$ is f -nef. Let $E = \sum E_i$ be the reduced exceptional divisor of f . Since X is log canonical, it follows that if m is sufficiently divisible then the $\mathbb{Q}$ -divisor

$$F := K_{m,Y/X} + E = K_Y + E - \frac{1}{m}A$$

is an effective exceptional $\mathbb{Q}$ -divisor. Let $N = \sum a_i E_i$ be characterized by $K_Y \equiv_f N$. We have

$$N + E - F \equiv_f \frac{1}{m}A.$$

In particular $-(N + E - F)$ is f -nef, and since it is exceptional, we conclude that $N + E - F \geq 0$ by the Negativity Lemma ([KM, Lemma 3.39]). This implies that $a_i \geq -1$ for all i , and hence that X is numerically log canonical. $\square$

Corollary 8.14. *Let X be a normal surface with log canonical singularities. Then X is $\mathbb{Q}$ -Gorenstein.*

Corollary 8.12 can be equivalently deduced from this corollary.

9. TERMINAL AND CANONICAL SINGULARITIES

In this section we deal with the generalization of canonical and terminal singularities, and discuss the corresponding extensions of invariance properties of singularities, plurigenera and numerical Kodaira dimensions established in the $\mathbb{Q}$ -Gorenstein case in works of Siu, Kawamata and Nakayama. Throughout the section, the ground field is assumed to be an algebraically closed field of characteristic zero.

Consider a pair (X, Z) , where X is a normal variety and Z is an effective formal linear combinations of proper closed subschemes of X .

Definition 9.1. The *log-discrepancy* of a prime divisor F over X with respect to (X, Z) is

$$a_F(X, Z) := \text{ord}_F(K_{X'/X}) + 1 - \text{val}_F(Z)$$

The pair (X, Z) is said to be *canonical* (resp., *terminal*) if $a_F(X, Z) \geq 1$ (resp., > 1) for every exceptional prime divisor F over X . We say that X is *canonical* (resp., *terminal*) if so is the log pair $(X, 0)$.

Of course these notions coincide with the familiar ones in the $\mathbb{Q}$ -Gorenstein case by Proposition 3.4. Canonical singularities admit the following useful characterization (which is well-known in the $\mathbb{Q}$ -Gorenstein case).

Proposition 9.2. *Let X be a normal variety, and suppose that $Z = \sum a_k \cdot Z_k$ is an effective formal $\mathbb{Q}$ -linear combination of effective Cartier divisors Z_k on X . Then the pair (X, Z) is canonical if and only if for all sufficiently divisible $m \geq 1$ (in particular, we ask that $ma_k \in \mathbb{Z}$ for every k), and for every log resolution $f: Y \rightarrow X$ of $(X, Z + \mathcal{O}(mK_X))$, there is an inclusion*

$$\mathcal{O}_X(m(K_X + Z)) \cdot \mathcal{O}_Y \subseteq \mathcal{O}_Y(m(K_Y + Z_Y))$$

as sub- $\mathcal{O}_Y$ -modules of $\mathcal{K}_Y$, where Z_Y is the proper transform of Z (as usual, the canonical divisors K_X and K_Y are chosen so that $f_*K_Y = K_X$).

Proof. Note that $f^{-1}(Z) = f^*Z = -f^*(-Z)$, once we think of Z and $f^{-1}(Z)$ as $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisors. If (X, Z) is canonical, and m and f are chosen as in the statement, then we see that

$$\begin{aligned} m(K_Y + Z_Y) + f^\natural(-m(K_X + Z)) &\geq m(K_Y + Z_Y) + f^*(-m(K_X - Z)) \\ &= m(K_Y + Z_Y + f^*(-K_X - Z)) = m(K_{Y/X} + Z_Y - f^{-1}(Z)) \geq 0 \end{aligned}$$

by Remark 2.14 and Proposition 2.13, and hence we get an inclusion as asserted. Conversely, suppose that (X, Z) is not canonical, and fix any log resolution $f: Y \rightarrow X$ of (X, Z) . By hypothesis, the $\mathbb{R}$ -divisor $K_Y + Z_Y + f^*(-K_X - Z)$ is not effective. Since $f^*(-K_X - Z)$ is the componentwise limit of the $\mathbb{Q}$ -divisors $\frac{1}{m} \cdot f^\natural(-m(K_X + Z))$, we can find a sufficiently large (and divisible) m such that $K_Y + Z_Y + \frac{1}{m} \cdot f^\natural(-m(K_X + Z))$ is not effective. By further blowing up, we may assume that f is a log resolution of $(X, Z + \mathcal{O}_X(mK_X))$. Then the assertion follows. $\square$

As an application, we show that deformation invariance of canonical singularities, plurigenera, and numerical Kodaira dimension also holds in this more general context. We first review the following (surely well-known) property.

Lemma 9.3. *If H is a normal irreducible Cartier divisor on a variety X , then X is normal in a neighborhood of H .*

Proof. Since H is Cartier and regular in codimension one, X is regular in codimension one near H . Let $f: X' \rightarrow X$ be the normalization. Over a neighborhood of H , f is an isomorphism away from sets of codimension ≥ 2 , hence the divisor $H' := f^*H$ is irreducible and reduced. Note also that H' is supported exactly on the inverse image of H . Note that $g = f|_{H'}: H' \rightarrow H$ is a finite birational surjective morphism of varieties. Since H is normal, $f|_{H'}$ is an isomorphism by Zariski Main Lemma. We have a commutative diagram with exact rows

$$\begin{array}{ccccccc} 0 & \longrightarrow & f_*\mathcal{O}_{X'}(-H') & \longrightarrow & f_*\mathcal{O}_{X'} & \longrightarrow & g_*\mathcal{O}_{H'} \longrightarrow 0 \\ & & \uparrow & & \uparrow & & \parallel \\ 0 & \longrightarrow & \mathcal{O}_X(-H) & \longrightarrow & \mathcal{O}_X & \longrightarrow & \mathcal{O}_H \longrightarrow 0. \end{array}$$

It follows that the cokernel of $\mathcal{O}_X \hookrightarrow f_*\mathcal{O}_{X'}$ is supported away from H , and hence that X is normal in a neighborhood of H . $\square$

We start with the extension of Kawamata's theorem on the deformation invariance of canonical singularities [Kaw2].

Theorem 9.4. *Let $f: X \rightarrow C$ be a flat morphism from a variety to a smooth curve such that, for some point $0 \in C$, the fiber $X_0 = f^{-1}(0)$ is a normal variety with only canonical singularities. Then (X, X_0) is canonical in a neighborhood of X_0 , and so are all fibers of f over a neighborhood of 0.*

Proof. The proof follows the arguments of [Kaw3]. By shrinking X near X_0 , we can assume that X is normal by Lemma 9.3. We may also assume that X_0 is affine. Let $m > 0$ be a sufficiently divisible integer and let $\mu: Y \rightarrow X$ be a log resolution of $(X, X_0 + \mathcal{O}_X(mK_X))$ which restricts to a log resolution of $(X_0, \mathcal{O}_{X_0}(mK_{X_0}))$. Let $\{s_1, \dots, s_k\}$ be a generating set of sections of $\mathcal{O}_{X_0}(mK_{X_0})$. Let Y_0 be the strict transform of X_0 . By Proposition 9.2, there is an inclusion

$$\mathcal{O}_{X_0}(mK_{X_0}) \cdot \mathcal{O}_Y \subseteq \mathcal{O}_{Y_0}(mK_{Y_0}).$$

One sees that there are corresponding sections $\tilde{s}_i$ of $\mathcal{O}_{Y_0}(mK_{Y_0})$ which push forward to the sections s_i of $\mathcal{O}_{X_0}(mK_{X_0})$. By Theorem A of [Kaw3], after possibly restricting over a neighborhood of P , these sections extend to sections $\tilde{S}_i$ of $\mathcal{O}_V(m(K_Y + Y_0))$. Pushing forward, we obtain sections S_i of $\mathcal{O}_X(m(K_X + X_0))$ that restrict to s_i . It follows that the S_i are generators of $\mathcal{O}_X(m(K_X + X_0))$ and that the inclusion $\mu_*\mathcal{O}_V(m(K_Y + Y_0)) \subseteq \mathcal{O}_X(m(K_X + X_0))$ is an equality. Therefore, there is an inclusion

$$\mathcal{O}_X(m(K_X + X_0)) \cdot \mathcal{O}_Y \subseteq \mathcal{O}_Y(m(K_Y + Y_0)),$$

and hence (X, X_0) is canonical. $\square$

Similarly, we have the following extension of the invariance of plurigenera (in the general type case) and of numerical Kodaira dimension for varieties with canonical singularities [Siu, Kaw2, Nak].

Theorem 9.5. *Let $f: X \rightarrow S$ be a projective flat morphism of varieties whose fibers $X_t = f^{-1}(t)$ are normal varieties with canonical singularities for every $t \in S$. Then the following properties hold:*

- (a) *The numerical Kodaira dimension $\nu(X_t)$ is constant on $t \in S$. In particular, if one fiber X_0 is of general type, then so are the other fibers.*
- (b) *Suppose additionally that the generic fiber X_η is a variety of general type. Then the plurigenera $P_m(X_t)$ is constant on $t \in S$ for any positive integer m .*

Proof. The proof is similar to those of Theorems 1.2' and 1.3 of [Kaw3], after we remark that Kodaira's lemma (cf. [Laz, Proposition 2.2.6]) holds for (not necessarily Cartier) divisors on a normal projective variety. $\square$

Remark 9.6. Canonical singularities on a $\mathbb{Q}$ -Gorenstein normal variety are obviously purely log terminal. However, it remains unclear whether an analogous implication still holds if the singularities are not $\mathbb{Q}$ -Gorenstein (cf. Remark 3.5).

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